



# Rounding errors

## Task A: Number lines

- 1) Work out the largest and smallest unrounded integers. You can use a number line if it helps.
 

|                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                       |
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| <p>a) 300 rounded to the nearest 10<br/> <i>295 is the smallest integer</i><br/> <i>304 is the largest integer</i></p> <p>b) 300 rounded to the nearest 100<br/> <i>250 is the smallest integer</i><br/> <i>349 is the largest integer</i></p> <p>c) 300 rounded to the nearest 50<br/> <i>275 is the smallest integer</i><br/> <i>324 is the largest integer</i></p> | <p>d) 640 rounded to the nearest 10<br/> <i>635 is the smallest integer</i><br/> <i>644 is the largest integer</i></p> <p>e) 3000 rounded to the nearest 100<br/> <i>2950 is the smallest integer</i><br/> <i>3049 is the largest integer</i></p> <p>f) 50 rounded to the nearest 50<br/> <i>25 is the smallest integer</i><br/> <i>74 is the largest integer</i></p> |
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- 2) Work out the unknown number. Draw number lines to help.
 

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| <p>a) 720 rounded to the nearest 10<br/> <i>715 is the smallest integer</i><br/> <i>724 is the largest integer</i></p> <p>b) 140 rounded to the nearest 20<br/> <i>130 is the smallest integer</i><br/> <i>149 is the largest integer</i></p> <p>c) 900 rounded to the nearest 50<br/> <i>875 is the smallest integer</i><br/> <i>924 is the largest integer</i></p> |  |
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- 3) In a school, the headteacher says there are 480 pupils rounded to the nearest 10. What is the largest number of pupils in the school?  

*The largest number of pupils in the school would be 484*
  
- 4) An advert says 56 000 people use their product rounded to the nearest 1000. What is the least amount of people who use their product?  

*The smallest number of people would be 55 500*



## Task B: The impact of rounding errors

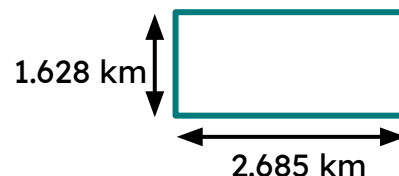
1) Here is a plot of land. Explain if and where students have lost accuracy and resulted in an incorrect answer when finding the area of the rectangle to 2 d.p.?

Aisha

$$1.628 \times 2.685 = 4.38 \text{ km}^2 \text{ (2 d.p.)}$$

$$1.628 \times 2.685 = 4.37118$$

Aisha has incorrectly rounded her answer



Jun

$$1.628 \times 2.685 = 4.37 \text{ km}^2 \text{ (2 d.p.)}$$

This is correctly rounded and Jun has used the accurate measurements given.

Andeep

$$1.63 \times 2.69 = 4.38 \text{ km}^2 \text{ (2 d.p.)}$$

Andeep has rounded the measurements before calculating the area. This has resulted in a loss of accuracy.

Laura

$$16.28 \times 26.85 = 437.118 \\ = 437.12 \text{ km}^2 \text{ (2 d.p.)}$$

Laura has incorrectly written her values. She has changed the place value of the digits and so her answer is far too big.

2) A circular sheet of metal has an area of  $25 \text{ m}^2$ . It gets two square patches with lengths  $2.6574 \text{ m}$  cut from it using a laser cutter.

Which pupil has correctly worked out the unpatched area to 2 d.p.? Explain the errors from the other pupils.

Aisha

$$\begin{aligned} \text{Area of squares:} \\ 2 \times 2.6574 \times 2.6574 \\ = 14.1 \text{ (1 d.p.)} \\ \text{Remaining area:} \\ 25 - 14.1 = \\ 10.90 \text{ m}^2 \text{ (2 d.p.)} \end{aligned}$$

Aisha has rounded too early.

She rounded her first calculation to 1 d.p. which has meant she has struggled to give the correct accurate answer to 2 d.p.

Lucas

$$\begin{aligned} \text{Area of squares:} \\ 2 \times 2.6574 \times 2.6574 \\ = 14.12 \text{ (2 d.p.)} \\ \text{Remaining area:} \\ 25 - 14.12 = \\ 10.88 \text{ m}^2 \text{ (2 d.p.)} \end{aligned}$$

Lucas has answered the question correctly.

He did round early, but stated it to 2 d.p. which resulted in less error loss.

Laura

$$\begin{aligned} \text{Area of squares:} \\ 2 \times 2.6574 \times 2.6574 \\ = 14.123 \dots \\ \text{Remaining area:} \\ 25 - 14.123 \dots = \\ 10.88 \text{ m}^2 \text{ (2 d.p.)} \end{aligned}$$

Laura has used the ellipsis to show she is using the ANS button on her calculator.

